Keyush Shah

+1 (267) 278-1269| keyush.shah12@gmail.com| LinkedIn | Website| Blog

EDUCATION

University of Pennsylvania (School of Engineering and Applied Science) — **Philadelphia, PA**
Master of Science in Engineering (MSE) in Data Science; ***GPA 3.9/4.0*** — ***Aug 2023 – May 2025***
Coursework: *Machine Learning, Databases, Computer Systems, Deep Learning, Generative Modeling, NLP, Machine Perception*

University of Delhi — **Delhi, India**
Bachelor of Science in Statistics; ***GPA 3.87/4.0*** — ***Aug 2017 – May 2020***

EXPERIENCE

Senior Software Engineer, Omnicom (NYC, New York) — ***Aug 2025 - Present***

- *Synthetic Audiences*:
  1. Architected a RAG-augmented product using **Next.js + FastAPI on Vercel/Railway** enabling pharma clients to simulate patient focus groups - interviewing AI-grounded personas & **reducing 2-4 weeks of research to minutes**.
  2. Developed a Go microservice fanning out **~100 concurrent LLM calls/request with bounded goroutines and context deadlines** over a JSON API decoupling serverless routes for ads/creative testing & better failure control.
- *Brand Intelligence Compiler (Knowledge Base):*
  1. Engineered a Claude based multi-step agent that orchestrated **routing->retrieval->tool-use** over a pharma knowledge base enforcing a **cited/abstain contract** + a self-verifying agent that lifted extraction by ~**60%**.
  2. Designed an offline **eval harness** monitoring a north star GRR + diagnostic LLM judge flagging quality regressions.
  3. Deployed **request-level telemetry** with async quality scoring, enabling label-free monitoring in production traffic.
- Led end-to-end migration of a multi-component ML Pipeline from AzureML to Databricks by re-architecting LLM Integration, data storage and maintaining orchestration in Spark Job DAGs unifying pipeline governance.
- Reduced CI/CD pipelines runtime by **12x** by benchmarking Docker containers & migrating to **UV for backend services.**

AI Researcher, *Penn Medicine* - *Computational Social Listening Lab* (PA, USA) — ***Dec 2024 – Aug 2026***
IH Risk Model (Part Time Research) **[CODE LINK]**

- Pioneered an ML hernia model using XGBoost, nested-CV and MLflow-tracked 21 ablations reducing model selection time and implementing SHAP based dilution corrections for feature importances and **raising AUC by 30% to 0.81**.
- Engineered a few shot LLM extraction pipeline on Azure OpenAI using a worker concurrency runner over ~1mm op notes (1.2M API calls, 9B tokens), hitting 96-97% field accuracy while **prompt caching cut cost by 48% ($31K to $16K).**
- Built a fault tolerant multi-agent orchestration system using Claude Agent SDK featuring **LLM-as-judge** observability, automated retry logic and **FastAPI + WebSocket web interface** slashing pipeline runtime from **45 to 10 min.**
- Integrated **vLLM** & **LangChain** for scalable local LLM inference deployed on A100 GPU and **modularized the backend** (OpenAI, Mistral) using **parallelized batch inference** and **memory-aware chunking** across variable length notes.

Data Science Intern (Full-time)**, *Universal Media*** (PA, USA) — ***May 2024 – Aug 2024***

- Led the development of **3+ data pipelines** using Azure Data Factory, facilitating seamless ingesting into Azure.
- Drove product insights by building **Mixed Media time series models**, analyzing channel impacts on media diversity metrics.

Assistant Manager (Full-time), ***IIFL Finance Ltd*** — ***Apr 2022 – July 2023***

- **Digital Adoption:** Led a product initiative to identify digitally savvy customers **–** engineered features & built ADF pipelines to track campaign behavior. Trained an ML model in AzureML and exposed as a REST endpoint driving disbursal by ~50%.

PUBLICATIONS

- **Learning Human-Perceived Fakeness in AI-Generated Videos via Multimodal LLMs –** COLM 2026 **[arXiv]**
  X. Fu, S. Liu, …, **K. Shah,** Y. Choi, J. Zou, D. Roth, C. Callison-Burch

SELECTED PROJECTS

- **Particle (2025):**
  - Built a context-aware chatbot for product and order support using FastAPI, LangChain, and Pinecone, implementing **RAG for product queries and tool-based logic with LLM fallback** for transactions. [Link]
  - Orchestrated a multi-turn memory enabled >95% accurate follow-up handling and improved support efficiency by 40%.

TECHNICAL SKILLS

Programming Languages: Python, Go, C/C++, SQL, JavaScript, Typescript, R programming
ML Libraries/Frameworks: PyTorch, scikit-learn, sentence-transformers, XGBoost, HuggingFace, spaCy, OpenCV, MLflow, A/B Testing
Databases/Web Frameworks: MySQL, PostgreSQL, SSMS, MongoDB, Neo4j, React, NodeJS, Django, Flask, FastAPI
Cloud/Big Data Orchestration: AWS (S3, Glue), Azure (Data Factory, Synapse), Vercel/Railway, Kafka, Airflow, DBT, PySpark
Tools/DevOps: DataBricks, Apache Spark, Docker, Jenkins, Git, Kubernetes, pytest, CI/CD